

6. (2.5 points)

Given the following information as of December 31, 2018:

Accident Year	On-Level Earned Premium (\$000,000)	Cumulative Paid Loss (\$000,000) as of (months)		
		12	24	36
2016	13,000	360	1,425	2,850
2017	13,250	375	1,375	
2018	13,500	350		

- The expected loss payment pattern follows a loglogistic curve of the form $\frac{x^\omega}{x^\omega + \theta^\omega}$, where
 - $\omega = 1.448$
 - $\theta = 48.021$
 - There are no payments after 120 months.
 - Accidents occur uniformly throughout the year.
 - The scale parameter, σ^2 , is 423.
- a. (1.75 points)

Calculate the incremental fitted payment and corresponding normalized residual for accident year 2018 at 12 months using the Cape Cod method.

b. (0.75 point)

Calculate ultimate losses for accident year 2016 using the Cape Cod method.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 6				
TOTAL POINT VALUE: 2.5			LEARNING OBJECTIVE(S): A2	
SAMPLE ANSWERS				
Part a: 1.75 points				
<u>Sample 1</u>				
$G(X) = \frac{x^\omega}{x^\omega + \theta^\omega}$		Used Prem = G(X) * Prem Ult = ELR * EP		
AY	X	G(X)	"Used Prem"	Ult
trunc	114	0.776		
2016	30	0.336	4,367.92	7,847.81
2017	18	0.195	2,577.45	7,998.73
2018	6	0.047	633.17	8,149.65
			7,578.54	
ELR = (2,850 + 1,375 + 350) / 7,578.54 = .604				
Expected Inc. _{2018,12} = .047(8149.65) = 382.23				
Norm Resid = $\frac{actual - expected}{\sqrt{\sigma^2 \times expected}}$ = $\frac{350 - 382.23}{\sqrt{423 \times 382.23}}$ = -.08				
<u>Sample 2</u>				
AY	Avg Age	G(X)	G(X)/0.776	"Used Prem"
trunc	114	0.776	1.0000	
2016	30	0.336	0.4321	5,617
2017	18	0.195	0.2501	3,314
2018	6	0.047	0.0603	814
				9,745
ELR = (2,850 + 1,375 + 350) / 9,745 = .4695				
Expected Inc. _{2018,12} = 13,500 x 0.4695 x .0603 = 382.2				
Norm Resid = $\frac{actual - expected}{\sqrt{\sigma^2 \times expected}}$ = $\frac{350 - 382.2}{\sqrt{423 \times 382.2}}$ = -.08				
Part b: 0.75 point				

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Sample 1

$$\begin{aligned}\text{Reserve}_{2016} &= \text{Ult} * [G(X_{\text{trunc}}) - G(X)] \\ &= 7,847.81(.776 - .336) = 3,465.84 \\ \text{Ult Loss}_{2016} &= 2,850 + 3,465.84 = 6,315.84\end{aligned}$$

Sample 2

$$\text{Ult Loss}_{2016} = 13,500 \times 0.4965 \times (1 - 0.4321) + 2,850 = 6,316$$

EXAMINER'S REPORT

Candidates were expected to be able to test how well the Cape Cod Method performed using a truncated development pattern by producing a normalized residual for the 2018 accident year. Candidates were also expected to produce an estimate of ultimate losses using the Cape Cod Method for the 2016 year.

Part a

Candidates were expected to produce a normalized residual using a truncated payment pattern.

Candidates were also expected to derive the payment pattern based on the given curve form and loss data and utilize that pattern to produce an overall estimated loss ratio (ELR). Using the overall ELR, candidates were expected to apply it to the 2018 earned premium along with the 2018 paid development pattern to produce the fitted incremental payment which is the underlying input to the normalized residual formula.

Common mistakes included:

- Applying a normalized truncated pattern to an ELR that was derived using a non-normalized pattern.
- Using the Cape Cod ultimate losses as the seed for the incremental payment calculation

Part b

Candidates were expected to produce ultimate losses for the 2016 accident year using the Cape Cod Method.

Common mistakes included:

- Producing ultimate losses for a year other than 2016
- Applying an incorrect incremental payment pattern
- Calculating the estimated ultimate losses as the ELR x Premium.